

Strength And Durability Properties of Sustainable Concrete Utilising Rice Husk Ash and Silica Fume

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Abstract. Supplemental cementitious material has been commonly added to concrete in recent years. By replacing some of the Ordinary Portland Cement (OPC) in concrete with cementitious elements like fly ash and silica fume, the mechanical properties of the mixture can be enhanced. There is an impending fly ash shortage in Arkansas because of nearby coal plant businesses. The possible use of rice husk ash (RHA) to replace cementitious materials has been investigated in this study. RHA is a waste product that is produced during agricultural processes. Due to its elevated silica content, it can be used as cementitious materials. In this investigation, concrete was modified to partially replace Ordinary Portland Cement using silica fume (0 to 20%) and RHA (0 to 30 %) at different percentages. In this investigation, a Class C fly ash that is often utilised locally was also included for comparison analysis with concrete specimens modified with RHA and silica fume. Both the fresh concrete mix's and hardened concrete's mechanical characteristics were ascertained through laboratory testing. The durability of the material will be improved, and the quantity of cement required will be reduced by the addition of silica fume and RHA to concrete. This includes improvements in water absorption, sorptivity, effective porosity, permeability, acid resistance, and resistance to fast chloride penetration. Nevertheless, the incorporation of silica fume and RHA in concrete leads to an elevation in water demands in comparison to the standard regular Portland cement mixture.

1 Introduction

Cement manufacture, especially the clinker formation process, releases a large amount of CO₂ into the atmosphere and uses a lot of energy, which exacerbates the greenhouse effect and contributes to global warming. As an additional cementitious ingredient, fly ash and granulated blast furnace slag, two byproducts of agricultural and industrial activities, help reduce environmental issues. A reduction in cement manufacturing leads to a drop in energy usage and carbon dioxide emissions. In addition to this, the mechanical qualities of concrete mixtures are improved by these chemicals, and the resistance to penetration is also increased. A significant portion of amorphous SiO₂ is present in siliceous materials

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including rice husk ash (RHA) and silica fume (SF). When these components are integrated into cement matrices, they exhibit chemical reactions that are analogous to one another. Cement-based products gain strength due to their heightened reactivity, resulting from pozzolanic interactions with calcium hydroxide produced during cement hydration.

The addition of silica fume and rice husk ash to concrete results in enhanced durability, increased strength, reduced permeability, and improved workability. To achieve these benefits, these materials can be included in concrete mixtures as partial replacements for cement. Because silica fume particles are so small, they can fill in the gaps left by cement particles, making concrete that is more compact and less porous. Conversely, rice husk ash includes active silica, which can enhance the overall effectiveness of concrete mixtures. Because they have both shown to increase the durability and life of concrete, rice husk ash and silica fume are useful additions to building projects. Moreover, the inclusion of these elements in concrete mixtures can further promote sustainable practices by making use of waste materials from other industries. This not only diminishes the ecological footprint of construction projects but also aids in the preservation of natural resources for future generations. In summary, incorporating silica fume and rice husk ash into concrete mixtures can result in the construction of more robust and long-lasting structures that are also eco-friendly. By integrating these materials, construction projects can simultaneously realise enhancements in performance and meet environmental objectives.

The by-product of smelting ferrosilicon or metallic silicon in an electric arc furnace is silica fume (SF), a finely divided non-crystalline silica used in the alloys industry. High-purity quartz undergoes a reduction to silicon at very high temperatures during the process, producing SiO_2 vapours that oxidise and condense into small particles that contain 85%–99% amorphous silica [1-4]. These particles have approximately spherical forms and typically have diameters between 20 and 500 nm. The estimated yearly production of SF worldwide is one to 1.5 million tonnes. SF-containing concrete is well known for being stronger and more resilient than regular Portland cement concrete, especially in sulfate-rich regions.

The residue that remains after burning rice husks the outer shells of paddy rice that are removed during husking is known as rice husk ash (RHA). One tonne of paddy rice burns to about 40 kg of ash. This ash is made when ortho silicic acid from groundwater is ingested by rice plants and polymerizes to form amorphous silica in the husks. Organic molecules burn to produce carbon dioxide, leaving silica behind in the ash[5]. The main component of RHA is silicon dioxide (SiO_2) generated from biomass, and the silica properties of the material are greatly influenced by the processing conditions. Cristobalite and tridymite, two less reactive crystalline silica minerals, are frequently found in ash from uncontrolled combustion, whereas burning RHA between 500 and 700 °C produces the largest amount of amorphous silica [6]. Because of its porous particle structure and high amorphous silica content, RHA has a wide surface area that contributes to its reactivity. Finer ash further enhances reactivity. RHA is a pozzolanic substance that can replace up to 15% of Portland cement; fine RHA, in particular, can reduce mortar porosity and improve compressive strength. RHA also increases the longevity of concrete by fortifying it against sulphate degradation, water permeability, and chloride ion penetration.

In recent years, there has been a substantial increase in the attention given to the utilisation of ecologically friendly components, such as rice husk ash (RHA) and silica fume (SF), in the production of concrete. In their study, Zareei et al. (2017) determined that the most effective amounts of SF and RHA to substitute cement in pastes and concretes were 10% by weight. The strength of the concretes was unaffected by the addition of up to 30% of SF or 15% of ground RHA, despite the fresh mixtures showing a noticeable reduction in slump. According to these results, concrete's mechanical properties can be positively impacted by using both SF and RHA as partial cement substitutes [7]. Xu et al.

(2016) presented further evidence supporting this finding by demonstrating that RHA may be utilised as a replacement for cement in concrete, within a range of 10% to 20%, without any adverse impact on its performance. The significant pozzolanic characteristics of RHA are responsible for this outcome. In addition, Bheel et al. (2020) found that the compressive strength of concrete containing rice husk ash (RHA) was higher than that of regular concrete, suggesting that RHA has a beneficial effect on the durability characteristics of concrete [4]. In addition, Sahoo et al. (2021) emphasised that incorporating SF and RHA into concrete resulted in a substantial improvement in its compressive strength, splitting tensile strength, and flexural strength. The researchers discovered that substituting 50% of SF led to an optimal blend, resulting in a notable enhancement in compressive strength and water absorption resistance when compared to the control combination. [6]. According to Das et al. (2020), the use of 10% total cementitious materials (5% RHA and 5% SF) resulted in a considerable enhancement of concrete's compressive strength, splitting tensile strength, and flexural strength by 12.65%, 9.40%, and 9.46% respectively after 28 days. These findings indicate that the simultaneous application of RHA and SF can result in substantial enhancements in the mechanical characteristics of concrete.[9]. Overall, the findings from the literature review suggest that concrete incorporating RHA and SF has the potential to meet the strength and durability requirements of construction materials while offering environmental benefits through the use of sustainable materials[10-14]. However, further research is needed to fully explore the optimal utilization of RHA and SF in concrete mixtures and their long-term performance in various environmental conditions.

2 Materials

2.1 Binding Materials

For the preparation of the specimens in this investigation, silica fume and regular Portland cement grade 43 were utilised. The cement's specific gravity, consistency, and first setting time were determined to be 3.215 and 31%, respectively. When compared to other concrete admixtures, silica fume is often regarded as one of the most effective ones for regular cement concrete. The objective of this experiment was to improve the compressive strength of concrete by substituting 0 to 20% of conventional Portland cement with silica fume.

In this study, Portland cement is replaced by silica fume and RHA. Table 1 displays the physical characteristics of the binders, as determined through an assessment conducted in compliance with the Indian Standard IS 4031. The experimental materials used in this study, silica fume and RHA, have larger particle sizes and higher specific surface areas compared to Portland cement, as indicated by their physical characteristics. Silica fume has a specific surface area of 200 m²/kg and a particle size of 10 µm, while RHA has a specific surface area of 170 m²/kg and a particle size of 25 µm. In contrast, Portland cement has a specific surface area of 320 m²/kg and a particle size ranging from 65 to 90 µm. Comparing these results to Portland cement, they show that both silica fume and RHA are highly refined and do not have a crystalline structure. Furthermore, compared to Portland cement, the specific gravities of the binders show that silica fume and RHA have lower weight densities.

Table 1. Physical properties of binding materials.

S.No	Property	OPC	SF	RHA
1	Specific Gravity	3.215	2.2	2.2
2	ParticalSize (micron)	65-90	10	25
3	Specific surface (m2/kg)	320	200	170
4	Initial Setting Time (min)	40	-	-
5	Final Setting Time (min)	520	-	-
6	Consistency	31%	+-	-

2.2 Fine Aggregate

The fine aggregate used was locally sourced sand that met the IS 383 zone II standards. It was carefully sieved using a 4.75 mm screen to ensure it was free from silt and organic debris. The specific gravity of sand, which is 2.65, was utilised. Table 2 provides information regarding the physical characteristics of the fine aggregate.

Table 2. Physical properties of Fine and Coarse Aggregate.

S.No	Property	Fine Aggregate	Coarse Aggregate
1	Water Absorption	1.52%	0.35%
2	Particle Shape	Rounded	Irregular
3	Fineness Modulus	2.65	6.35
4	Specific Gravity	2.65	2.657

2.3 Coarse Aggregate

The natural coarse aggregate has a specific gravity of 2.657 and a maximum size of 12.5 mm. It passes through a 4.75 mm screen and is retained. Table 2 shows the physical attributes of the natural coarse aggregate.

3 Concrete mix proportions

In this experiment, an attempt is made to produce high strength concrete M30 utilising silica fume and RHA. Six distinct mixes—one of which serves as a control—are created for experimental investigations. RHA is intended to replace Portland cement in other concrete mixtures by 30%. The group designed four concrete mixes by replacing 30% of silica fume with RHA and substituting 5%, 10%, 15%, and 20% of Portland cement. As a result, the aggregates were assumed to be surface saturated dry for calculating the material quantities. Table 3 displays the specifics of the material quantities determined for each of the six concrete mixes along with their mix names. Once the concrete was poured into the moulds, it was thoroughly compacted using a table vibrator. After being removed from the mould, the specimens were placed in a curing tank for the purpose of curing. The investigation involved casting specimens for the compression test, flexure test, and split tensile strength tests, as seen in Figure 1. Subsequently, these specimens were kept for a duration of 28

days in order to undergo the process of curing. Following this period, their individual strengths were assessed. The durability characteristics of high-strength concrete containing silica fume and RHA are evaluated by examining and comparing the specific water sorptivity and chloride particle penetration. [15-17].



Fig. 1. Casting and curing of specimen.

Table 3. Mix proportion for normal ordinary portland cement.

Mix Id	Cement kg/m ³	Silica fume kg/m ³	RHA kg/m ³	Sand kg/m ³	Natural Aggregate kg/m ³	Water kg/m ³
M1	400	-	-	762.244	984.68	160
M2	280	-	120	762.244	984.68	160
M3	260	20	120	762.244	984.68	160
M4	240	40	120	762.244	984.68	160
M5	220	60	120	762.244	984.68	160
M6	200	80	120	762.244	984.68	160

4 Result and discussion

4.1 Compressive Strength

The Figure 2 demonstrates the significant impact of substituting silica fume and RHA on the compressive strength of concrete at 7 and 28 days. After a period of 28 days, the compressive strength of M1 was measured to be 31.6 N/mm². In comparison, the concrete mixes for M2, M3, M4, M5, and M6, which contained different percentages of silica fume and RHA, had compressive strengths of 33 N/mm², 34 N/mm², 38 N/mm², 42 N/mm², and 40 N/mm² respectively. After thorough examination, it was seen that the compressive strength of the specimens rose proportionally with the level of substitution of the binding substance. On the other hand, there was an increase in compressive strength when thirty percent of the binding material was substituted with RHA. It was discovered that the compressive strength of every single concrete mix was higher than the specified strength value, with the exception of the mixes that had considerable amounts of silica components..

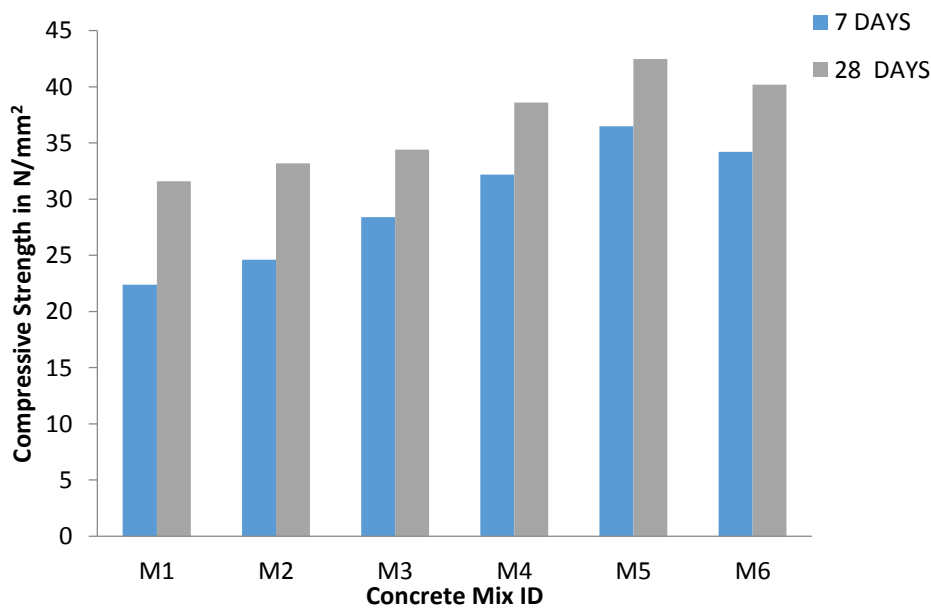


Fig. 2. Compressive Strength of Concrete

4.2 Split Tensile Strength

Figure 3 result shows that adding more replacement binding elements with a high silica content to the concrete mix enhanced the fraction of splitting tensile strength. High strength concrete's pozzolanic action is often increased by adding silica fume. It prompted us to improve each and every mechanical characteristic of high-strength concrete.

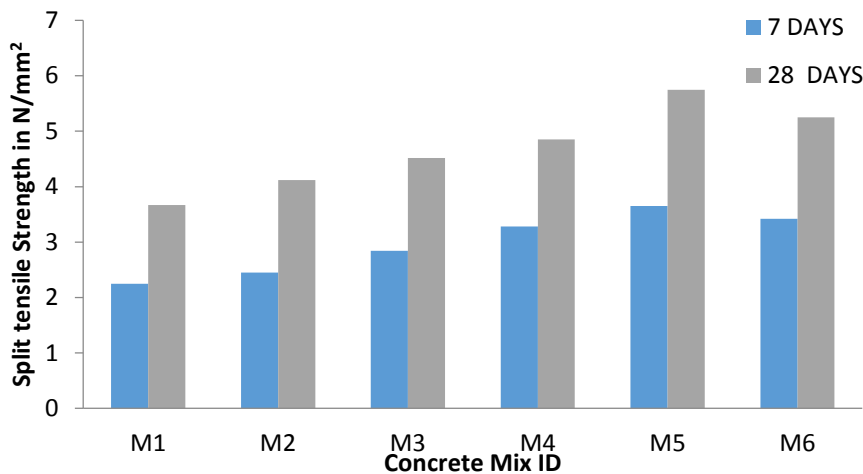


Fig. 3. Split Tensile Strength of Concrete

4.3 Flexural Strength

Figure 4 displays the flexural strength over a period of 28 days for different levels of substitution of silica fume and RHA. These data indicate that the level of strength enhancement in both Silica fume based and rice husk ash -based concrete increases as time progresses. This demonstrates a gradual reaction involving pozzolanic materials, and the obtained results closely align with the findings of an initial literature review. When the effectiveness of Silica fume and rice husk ash increasing strength at various ages is compared, it becomes clear that the concrete mixes based on SF exhibit superior strength increase at 28 days. With a longer curing time, RHA increases the cement's rate of hydration in the concrete. It is discovered that at later ages, the strengths of concrete are enhanced similarly by the actions of SF and RHA.

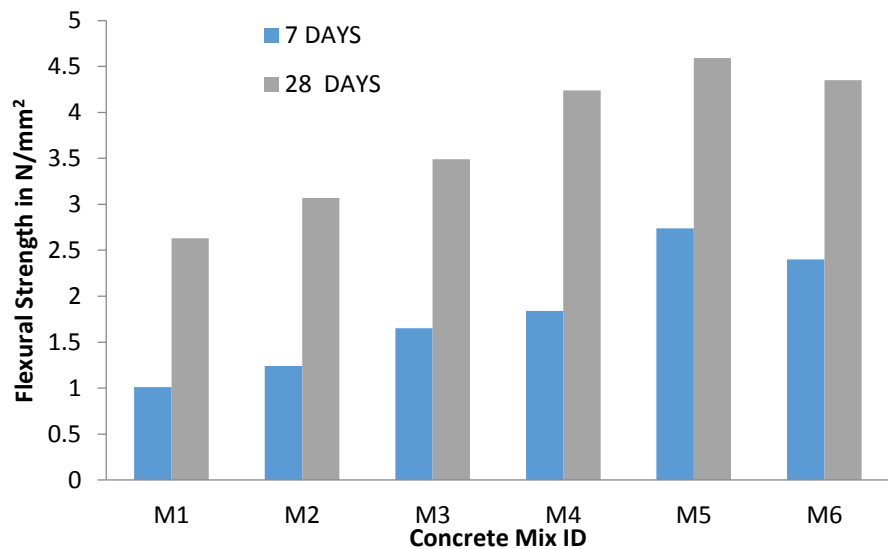


Fig. 4. Flexural Strength of Concrete

4.4 Rapid Chloride Permeability Test

The Rapid Chloride Permeability Test (RCPT) is conducted on a solid sample, as shown in Figure 5, which is either 50 mm or 100 mm thick. The test involves applying a 60 V DC voltage for a duration of six hours, following the guidelines of the ASTM C1202 test. Figure 4 depicts the arrangement of a 3.0% NaCl solution in one container and a 0.3 M NaOH solution in the other container. The experiment's outcomes are presented in Table 4. The concrete specimen utilised in this experiment was obtained during a 28-day period of curing, and the presence of chloride passing through it serves as an indicator of the material's permeability. The electric current is quantified by the equipment's showing, and these quantifications are subsequently compared to the mandated standard values set by ASTM C1202.



Fig. 5. Rapid Chloride Permeability Test

Table 4. Rapid Chloride Permeability.

S.No	Mix Id	Charge Pass In Coloumbs (C)	As Per ASTM
1	M1	2602	Moderate
2	M2	2402	Moderate
3	M3	2212	Moderate
4	M4	2102	Moderate
5	M5	2024	Moderate
6	M6	1815	LOW

4.5 Sorptivity

The extent of water infiltration is employed to assess the water permeability of concrete blends. The study observed variations in water infiltration depth among different concrete

mixes at different stages of concrete maturity in shown fig.6. At the 28-day mark, it is seen that the disparity in the depth of penetration for concrete mixes utilising SF and RHA is quite comparable. Regardless of age, the depth of water infiltration reduces as the content of SF or RHA increases. The reason for the decrease in penetration depth is the existence of tiny particles in SF and RHA, which block the concrete's pores. On a microscopic level, the density of the concrete mixes is increased as a result of the pozzolanic nature of SF/RHA, which leads to the production of extra C-S-H within the mixture. The quantity of water that the concrete specimen has absorbed through capillary action is measured by its sorptivity, which is shown in table 5. In the concrete specimen's pore structure, it shows a negative association with the degree of compactness. Because these two characteristics and the pore structure of concrete are strongly correlated, it has been discovered that the water permeability and the variation in sorptivity, or the S value, of concrete mixes that contain either SF or RHA are very similar.



Fig. 6. Sorptivity

Table 5. Sorptivity Test.

S.No	Mix Id	Sorptivity x 10 ⁻⁶ mm/ min0.5	Rate of Absorption I mm
1	M1	7.36	168.24
2	M2	6.72	148.25
3	M3	6.58	134.57
4	M4	5.48	125.60
5	M5	4.32	114.45
6	M6	3.82	102.28

5 Conclusion

The According to the preliminary research on the effects of RHA in high strength concrete with silica fume, the ensuing end may be suffocating.

- The addition of silica fume to RHA concrete increases its flexural and compressive strengths. 20% silica fume replacement lowers the compressive strength.
- The concrete matrix becomes denser as a result of silica fume, which increases its ability to support loads and prevents cracking when loads are applied.
- The concrete matrix's pores are filled with silica fume, which lowers the material's permeability to water and harsh chemicals.
- RHA concrete is more durable as a result of its reduced permeability, particularly in areas where moisture intrusion or chemical attacks are common.
- The interaction between silica fume and calcium hydroxide through the pozzolanic reaction strengthens the concrete structure, enhancing its durability against weathering and chemical corrosion.

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